Janjanam Kedarananda

ee24btech11030@iith.ac.in | kedaranandajanjanam@gmail.com | +91 7670996990 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.18
Lakshya School Samalkota Mandal East Godavari District Andhra Pradesh	Class XII (CBSE)	2024	94.00%
Aditya High School Kakinada	Class X (BSEAP)	2022	96.17%

EXPERIENCE

Texas Instruments (BYTE Certified) :

- Qualified the Texas Instruments BYTE certification assessment and was shortlisted among the top 3 participants.

Core Member | Elektronica - Electronics and Signal Processing Club, IITH :

- Delivered hands-on sessions on FPGA as a part of spring boot-camp, collaborated in RISC-V Processor design and FPGA implementation

Teaching assisant | Matrix Theory - EE1030 :

- Mentored students on mini-projects, evaluated assignments, and conducted tutorials on Python, LaTeX, and GitHub.

PROJECTS

Circuit Design in SPICE :

- CMOS inverter for equal rise and fall times; evaluated propagation delay with and without a 10 pF C_load and characterized its VTC and I-V.
- CMOS NAND and NOR gates for equal rise and fall times; evaluated propagation delay with and without a 10 pF C_load.
- Common source amplifier of small signal gain 25 with DC bias current of 0.1 mA.
- Diode-Op-amp based single stage half wave and two stage full wave precision rectifiers for accurate AC signal rectification.
- Schmitt trigger with Vth = 200mv; Square wave, triangular wave and sawtooth wave generator with target frequency 10KHz.
- Inverting and non inverting ideal and practical integrator and differentiator circuits using Op-amp.

Electronic Devices and Circuits :

- Diode DC/AC characteristics, extracting n, Is, and rd, and verified small signal and nonlinear behavior.
- Analyzed NMOS Id vs Vds/Vgs characteristics across cutoff, triode, and saturation regions.
- Characterized MOSFET small signal parameters gm, r0, gmb using analytical calculations and simulation across selected bias points

Custom gaming controller PCB :

- Developed firmware for ESP32 BLE controller integrating joysticks, D-pad, triggers, motion sensing, and LEDs
- Implemented joystick calibration, deadzone compensation, axis mapping, input smoothing, and gyroscope fusion to improve control accuracy and responsiveness.

RISC-V 32-64 bit Processor Design :

- Built a 5-stage pipelined RISC-V (RV32I/RV64I) processor in Verilog with IF, ID, EX, MEM, and WB stages

8-bit Breadboard Computer :

- Built an 8-bit computer integrating datapath (ALU, registers, RAM, Program Counter) and control logic (clock and microcode controller).
- Validated microcode sequencing, ALU operations, register transfers, memory access, clock timing, and shared bus functionality.

6-bit SAR ADC :

- Build a 6-bit SAR ADC on breadboard using Arduino-based SAR controller, Sample-and-Hold circuit, R-2R ladder DAC, and comparator.
- Characterized ADC performance using the following metrics -

DNL: 0.897 to 0.194 LSB, INL: 2.032 to 0.500 LSB, offset: 0.397 LSB, gain: 0.224 LSB and experimentally verified the Nyquist theorem using sampling frequency of 159.5 Hz, demonstrating 5 Hz reconstruction and aliasing at 184.07 Hz and 275 Hz.

SKILLS

Microcontrollers & FPGA	Arduino | ESP32 | Pynq-Z2 | ZCU-104	**Design & Simulation**	KiCad | LTspice | Cadence Virtuoso| Vivado
Programming	C | Python | Verilog | Bash	**Development Tools**	Git | GitHub | Linux | Peta-Linux | VS Code

SCHOLASTIC ACHIEVEMENTS

- Awarded a 1,00,000 LPUNEST 2024 study grant for securing a rank among the Top 500

RELEVANT COURSES

VLSI Courses : Circuits and Network Analysis, Control Systems, Analog Electronics, Introduction to CMOS Processing, Semiconductor Device Fundamentals, Power System Analysis, Signal Processing, Digital Systems, Computer Architecture, Digital Sytems Lab, Analog Lab, Electrical Circuits Lab, Electronic Devices and Circuits Lab, Electronic System Design Project Lab (PCB Design)

EXTRACURRICULAR

- Member of the Institute Hockey Team representing the institute in hockey leagues and tournaments.
- Participated in CeNSE Winter School on Semiconductor Technology conducted by IISC, Bengaluru

POSITIONS OF RESPONSIBILITY

- **Head | Hospitality & Logistics, SPIC MACAY, IITH :** Organized 5+ cultural events and managed accommodation, transportation, and hospitality for distinguished guests, including Padma Shri and Padma Bhushan awardees.